



Riverside OS v0.70.2 Printable Deployment Guide

Authoritative reference manual for store hardware configuration, software package updates, schema migrations, and backup verification.

TARGET VERSION	v0.70.2
DOCUMENT CLASSIFICATION	Operational Standard Work
AUTHORIZED STATIONS	Server PC / Register #1 / Register #2 (iPad) / Counterpoint PC
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Use this guide for a live-store install or recovery pass. It is written in the order the work should happen:

1. Confirm release files and store network facts.
2. Install and verify the Backoffice / Server PC.
3. Configure integration credentials and Counterpoint bridge access.
4. Install Register #1.
5. Install the iPad PWA lane.
6. Validate printers, scanners, sales flow, backup, and rollback.

Print this guide and check off each section during deployment.

1. Deployment Model

Store Roles

Station	Device	What it does
Backoffice / Server PC	Windows PC	Runs PostgreSQL, Riverside Server API, web/PWA files, migrations, firewall rule, backups, and the startup task. Keep this PC on.
Register #1	Windows PC	Main cashier station. Owns the receipt printer, cash drawer, and till group close.
Register #2	iPad PWA	Satellite register lane. Uses the server over the store network and attaches to the Register #1 till group.
Other Back Office PCs	Browser/PWA or desktop app	Back-office work against the same server. Do not install the server role here.
Counterpoint PC	Existing Windows/SQL host	Runs the Counterpoint bridge only when importing or syncing legacy Counterpoint data.

API Host Rule

Use the right API host before troubleshooting anything else.

Device	API host value
Backoffice / Server PC	http://127.0.0.1:3000
Register #1 Windows PC	http://SERVER-IP:3000, for example http://10.64.70.196:3000
Register #2 iPad PWA	http://SERVER-IP:3000, for example http://10.64.70.196:3000
Other Back Office PCs	http://SERVER-IP:3000 or trusted HTTPS/Tailscale URL
Counterpoint bridge .env	R0S_BASE_URL=http://SERVER-IP:3000

Do not use `127.0.0.1` on Register #1, iPad, or the Counterpoint bridge unless Riverside Server is running on that same device.

Register Cash Rule

- Register #1 is the physical cash drawer lane.
- Register #2 and other satellite lanes may enter sales, but physical cash still belongs to Register #1.
- Register #1 performs the till group close / Z-close.

2. Release Files And Records

Use only the official GitHub release files for the main install:

<https://github.com/cpg716/riverside-os/releases/tag/v0.70.2>

Required release assets:

- ☐ `RiversideOS-v0.70.2-Windows-Deployment.zip`
- ☐ `Riverside.POS_0.70.2_x64_en-US.msi`
- ☐ `Riverside.POS_0.70.2_x64_en-US.msi.sig`
- ☐ `latest.json`
- ☐ `riverside-updater-build-manifest.json`

Expected release identity:

Version: `0.70.2`
 Deployment package: `RiversideOS-v0.70.2-Windows-Deployment.zip`

Before starting:

- ☐ Unzip the Windows deployment package.
- ☐ Keep the package available on the Backoffice / Server PC and Register #1.
- ☐ Confirm Windows Administrator access on the Backoffice / Server PC.
- ☐ Confirm Windows Administrator access on Register #1.
- ☐ Record the Backoffice / Server PC name and LAN IP.
- ☐ Confirm Register #1, iPad, and Counterpoint PC can reach the same store network.
- ☐ Confirm Epson receipt printer IP or Windows printer name.
- ☐ Confirm scanner is in HID keyboard mode.
- ☐ Confirm payment terminal device codes and provider credentials if using Helcim.
- ☐ Keep the prior installer/build and a fresh database backup available for rollback.

Recommended:

- ☐ Put the Backoffice / Server PC on UPS power.
- ☐ Reserve static/DHCP IPs for the server PC and receipt printer.
- ☐ Record every generated password, station role, device IP, and install result in the deployment log.

3. Backoffice / Server PC

Install the server first. Nothing else should be installed until this machine is healthy.

3.1 Install

1. On the Backoffice / Server PC, open the unzipped deployment package.
2. Run:

```
Start-RiversideDeployment.cmd
```

3. Choose:

```
Backoffice / Server
```

4. Click **Check**.
5. Resolve failed readiness checks.
6. Click **Install**.

The installer should:

- Install or use PostgreSQL.
- Create/update the Riverside database user.
- Create/update the Riverside database.
- Copy `riverside-server.exe` .
- Copy web/PWA files.
- Copy migrations and seed files.
- Apply migrations.
- Download the pinned ROSIE LLM model (Gemma E4B, ~5.4 GB) and install the sherpa-onnx Python voice stack.
- Write `C:\RiversideOS\server\.env` .
- Write server credential secrets into the Windows machine environment.
- Add the inbound firewall rule for port `3000` .
- Create the scheduled task `Riverside OS Server` .
- Start Riverside Server.
- Check `http://127.0.0.1:3000/api/staff/list-for-pos` .

3.2 Password And Secret Notes

- If PostgreSQL already exists, enter the existing `postgres` admin password.
- If the manager installs PostgreSQL and password fields are blank, it can generate passwords.
- Riverside database password is generated when blank or placeholder.
- Riverside app secret is generated when blank, too short, or placeholder.
- `RIVERSIDE_CREDENTIALS_KEY` is required before Settings can save encrypted integration credentials.
- Helcim, Shippo, QBO, Podium, Counterpoint, and other provider credentials should be entered in Backoffice Settings after the server is healthy.
- Do not use local `.env` edits as the normal way to configure integrations. Use the Windows repair tools only when the server credential key or Counterpoint bridge token needs recovery.
- Do not commit or publicly share `riverside-deployment.config.json` .

3.3 Server Acceptance

On the Backoffice / Server PC:

- ☐ Riverside opens locally.
- ☐ Staff sign-in screen loads.
- ☐ API host is `http://127.0.0.1:3000` .

- ☐ Settings → Updates shows Riverside 0.70.2 .
- ☐ Settings → General → About this build shows the expected server/API.
- ☐ Settings → Data & Backups opens.
- ☐ Backup path is writable.
- ☐ Server restart works through the Deployment Manager or the Riverside OS Server scheduled task.
- ☐ Deployment Manager PostgreSQL Status panel shows service **running**, connection **OK**, and the riverside_os database exists.

From another device on the store network:

- ☐ Open http://SERVER-IP:3000 .
- ☐ Staff sign-in screen loads.
- ☐ No stale/update mismatch warning appears.

3.4 Server Repair Tools

Use these only when the symptom matches.

Tool	Use when
Deployment Manager Repair	Server files, firewall, scheduled task, migrations, or .env need to be refreshed without resetting the database.
Deployment Manager PostgreSQL Status	Check PG service state, connectivity, version, DB size, and migration count. Use Start PG / Restart PG / Stop PG buttons to control the service.
Deployment Manager Stop Server	Cleanly stop the Riverside OS Server task and process.
Deployment Manager Uninstall Server	Remove the server binary, task, and firewall rule while preserving the database.
Install-RosieAiStack.cmd	Installs the ROSIE AI models and Python voice tools, patches the server .env, and restarts the server without a full reinstall.
Repair-RiversideCredentialsKey.cmd	Settings says RIVERSIDE_CREDENTIALS_KEY must be set before integration credentials can be saved.
Set-CounterpointBridgeToken.cmd	Counterpoint bridge reaches Riverside but fails with health 401.
Reset-RiversideDatabase.cmd	Fresh pre-live install is badly interrupted and you intentionally need to reset the Riverside database. Do not use on live production data without rollback approval.

4. Integration Credentials And Counterpoint

Do this after the server is installed and before relying on Helcim, QBO, Counterpoint, Podium, Meilisearch, or other integration credentials.

4.1 Credential Storage Check

In Riverside:

1. Sign in as an admin.
2. Open **Settings**.
3. Open the integration you need.
4. Save the required credential.

If a warning says `RIVERSIDE_CREDENTIALS_KEY` must be set :

1. On the Backoffice / Server PC, open the deployment package.
2. Run:

```
Repair-RiversideCredentialsKey.cmd
```

3. Let it elevate and restart Riverside Server.
4. Close and reopen Riverside.
5. Save the credential again.

4.2 Counterpoint Bridge Setup

Use this section only if the store is running the Counterpoint bridge.

In Riverside:

1. Open **Settings → Counterpoint → Connect & Run**.
2. Save the **Bridge sync token**.

On the Counterpoint PC:

1. Open:

```
C:\counterpoint-bridge\.env
```

2. Set the Riverside server URL:

```
ROS_BASE_URL=http://SERVER-IP:3000
```

Example:

```
ROS_BASE_URL=http://10.64.70.196:3000
```

3. Set the same bridge token:

```
COUNTERPOINT_SYNC_TOKEN=the-exact-same-token
```

4. Save the file.

5. Restart the Counterpoint bridge.

4.3 Counterpoint Error Meanings

Bridge error	Meaning	Fix
health 503	Riverside is reachable, but Riverside Server does not have COUNTERPOINT_SYNC_TOKEN configured.	Save the token in Settings or run Repair-RiversideCredentialsKey.cmd on the server PC.
health 401	Riverside has a token, but the bridge sent a different token or no token.	Run Set-CounterpointBridgeToken.cmd on the server PC and paste the exact token from C:\counterpoint-bridge\.env, then restart the bridge.
Connection refused	The bridge cannot reach Riverside Server at ROS_BASE_URL.	Confirm server is running, URL uses port 3000, and firewall allows the connection.
SQL login/object errors	Bridge reached Counterpoint SQL but query/login/database is wrong.	Check SQL_CONNECTION_STRING and confirm Database= is the Counterpoint company database.

4.4 Search, Help, And ROSIE

After the server is stable:

- ☐ If Meilisearch is enabled, open Settings → Integrations → Meilisearch.
- ☐ Rebuild the search index after deploy, restore, Meilisearch wipe, or major import.
- ☐ Confirm product/customer/order search returns expected matches.
- ☐ Confirm Help Center opens.

- ☐ Confirm ROSIE only if the local runtime/model is installed and configured.

5. Register #1 Windows PC

Install Register #1 after the Backoffice / Server PC passes acceptance.

5.1 Install

1. On Register #1, open the unzipped deployment package.
2. Run:

```
Start-RiversideDeployment.cmd
```

3. Choose:

```
Register #1
```

4. Enter the Backoffice / Server PC API base.

Correct examples:

```
http://SERVER-PC-NAME:3000  
http://10.64.70.196:3000
```

Do not use:

```
http://127.0.0.1:3000
```

unless Riverside Server is running on Register #1 itself.

5. Enter Register #1 station label.
6. Choose receipt printer mode and cash drawer setting.
7. Click **Check**.
8. Resolve failed checks.
9. Click **Install**.

The installer should:

- Write `C:\ProgramData\RiversideOS\station-config.json`.
- Install the Riverside desktop app.

- Save the API base.
- Save printer and cash drawer settings.
- Launch the app when possible.

5.2 First Launch

- ☐ Open the Riverside desktop app.
- ☐ Confirm the app icon is the Riverside logo, not an old placeholder.
- ☐ Confirm Settings → Updates shows `0.70.2`.
- ☐ Confirm Settings → General → About this build shows `http://SERVER-IP:3000`.
- ☐ Staff sign-in works.
- ☐ Open Register #1.
- ☐ Enter the opening float according to store procedure.
- ☐ Confirm Register #1 is the cash drawer lane.

6. Receipt Printer, Cash Drawer, And Scanners

Do hardware validation on Register #1 before iPad/PWA validation.

6.1 Receipt Printer

Preferred receipt printer setup:

```
Mode: Network address
Printer: Epson TM-m30III
Port: 9100
IP: static or DHCP-reserved Epson IP
```

Use **Installed printer on this PC** only for USB/driver-managed fallback.

PWA receipt printing is still direct hardware printing. The iPad/PWA sends the request to Riverside Server, and Riverside Server opens the Epson printer IP/port. It should not open the browser print dialog for receipts.

Checklist:

- ☐ Epson receipt printer has paper.
- ☐ Epson printer IP or Windows printer name is recorded.
- ☐ Printers & Scanners settings are saved.
- ☐ Test receipt prints.

- ☐ PWA receipt test prints from the configured Epson IP/port.
- ☐ Register #1 receipt reprint path works.

6.2 Cash Drawer

- ☐ Cash drawer is connected to the receipt printer.
- ☐ Cash drawer opens for CASH/CHECK tender.
- ☐ Cash drawer does not intentionally open for card-only tender or receipt reprint.
- ☐ Staff understand physical cash belongs to Register #1.

6.3 Scanner

- ☐ Scanner is in HID keyboard mode.
- ☐ Scanner sends Enter/Tab suffix if desired.
- ☐ POS product search receives scanned text.
- ☐ Inventory/search input receives scanned text.
- ☐ Charging/spare scanner plan is known.

6.4 Reports And Tags

- ☐ Report printing uses browser/Windows print and the selected report printer.
- ☐ Clothing tags use the Zebra 2844 station by installed printer name or network IP.
- ☐ Zebra network checks fail clearly if Riverside Server cannot reach the tag printer.

7. Register #2 iPad PWA

Register #2 is a satellite lane. It uses the Backoffice / Server PC URL while inside the store.

7.1 Prepare iPad

- ☐ iPad is on the same store Wi-Fi/network as the Backoffice / Server PC.
- ☐ iPad date/time is correct.
- ☐ Safari is available.
- ☐ Bluetooth scanner, if used, is paired as keyboard/HID input.

7.2 Install PWA

1. Open Safari.
2. Visit the Riverside server URL.

Examples:

```
http://SERVER-PC-NAME:3000  
http://10.64.70.196:3000
```

3. Confirm the Riverside sign-in screen loads.

4. Tap **Share**.

5. Tap **Add to Home Screen**.

6. Name it:

Riverside Register 2

7. Launch Riverside from the new Home Screen icon.

Do not run Register #2 from a regular Safari tab for production use.

7.3 First Launch

- ☐ Launch from the Home Screen icon.
- ☐ Staff sign-in works.
- ☐ Settings → Updates shows 0.70.2 .
- ☐ No **Update incomplete** warning appears.
- ☐ Open/attach Register #2 only after Register #1 is open.
- ☐ Confirm it behaves as a satellite lane.
- ☐ Confirm customer lookup works.
- ☐ Confirm item lookup/cart works.
- ☐ Confirm checkout drawer fits comfortably on the iPad.
- ☐ Confirm staff know cash is handled at Register #1.

7.4 iPad Scanner And Receipt Path

- ☐ Pair Zebra CS6080 or similar as Bluetooth keyboard/HID.
- ☐ Open a product/SKU search field.
- ☐ Scan a barcode.
- ☐ Confirm scanned text appears in the focused field.
- ☐ If iPad receipt printing is expected, confirm the server can reach the network printer.
- ☐ If iPad print fails, reprint from Register #1.

8. Other Back Office Devices

For Windows PCs that are not the server and not Register #1:

- ☐ Open the Riverside server URL in Edge/Chrome.
- ☐ Use browser install prompt if desired.
- ☐ Sign in with staff PIN.
- ☐ Confirm Customers loads.
- ☐ Confirm Orders loads.
- ☐ Confirm Inventory loads.
- ☐ Confirm Reports/Insights loads if the role has access.
- ☐ Confirm Settings → Updates shows 0.70.2 .
- ☐ Do not install the Backoffice / Server role on these PCs.

Use the Riverside desktop app only when native printer/scanner reliability is required.

9. Production Smoke Test

Run this before first customer use.

Server

- ☐ Reboot or restart the Backoffice / Server PC.
- ☐ Confirm Riverside OS Server starts automatically.
- ☐ Confirm sign-in works on the server PC.
- ☐ Confirm another device can open http://SERVER-IP:3000 .
- ☐ Confirm backup path is writable.
- ☐ Deployment Manager PostgreSQL Status shows **running** and **OK**.

Register #1

- ☐ Staff sign-in works.
- ☐ Register #1 opens.
- ☐ Scanner works.
- ☐ Receipt prints.
- ☐ Cash drawer opens for CASH/CHECK.
- ☐ Card/Helcim flow is validated if enabled.
- ☐ Z-close path is understood.

Register #2 iPad

- ☐ Launches from Home Screen icon.
- ☐ Staff sign-in works.
- ☐ Register #2 attaches after Register #1 is open.
- ☐ Scanner works if used.
- ☐ Receipt fallback path is understood.
- ☐ No stale version warning.

Safe Sale Flow

Run one supervised low-risk flow:

- ☐ Sign in as staff.
- ☐ Open Register #1.
- ☐ Search or scan an item.
- ☐ Add item to cart.
- ☐ Open checkout drawer.
- ☐ Cancel safely or complete a store-approved test sale.
- ☐ Confirm receipt behavior.
- ☐ Confirm drawer behavior.
- ☐ Confirm the transaction appears in transaction history if completed.

10. Updates And Hotfixes

After any update:

- ☐ Update Backoffice / Server PC first.
- ☐ Confirm server starts and Settings → Updates shows 0.70.2 .
- ☐ Update Register #1 desktop app.
- ☐ Relaunch Register #1 and confirm Settings → Updates shows 0.70.2 .
- ☐ On iPad, open Settings → Updates → iPad and browser app → Check app files.
- ☐ If the PWA update prompt appears, tap **Reload now** only when no sale is in progress.
- ☐ Confirm no station shows **Update incomplete**.

Never run sales if the server, Windows app, and PWA files show different Riverside versions.

Update package rule:

- Use the latest provided deployment zip for the specific version.
 - Prefer the included `.cmd` helpers for targeted repairs.
 - Do not reset the database as a troubleshooting shortcut.
-

11. Backup And Rollback

Before live use:

- ☐ Create a fresh database backup.
- ☐ Confirm the backup file exists and has non-trivial size.
- ☐ Keep a copy off the server PC if practical.
- ☐ Keep the previous installer/build accessible.
- ☐ Record backup timestamp and file name.
- ☐ Record who has rollback authority.

Rollback rule:

1. Stop all registers.
 2. Stop active checkouts.
 3. Record the reason for rollback.
 4. Restore the database backup before running older binaries against migrated data.
 5. Do not run an older server against a newer migrated database without restore approval.
-

12. Quick Troubleshooting

PWA will not load

1. Confirm the device is on the correct network.
2. Open `http://SERVER-IP:3000` in the device browser.
3. Check DNS/server name.
4. Check firewall on the Backoffice / Server PC.
5. If using HTTPS, check certificate/date/time.
6. Close and reopen the Home Screen icon.
7. If still stale, remove the icon, clear site data, and Add to Home Screen again.

Register app will not reach server

1. Check Settings → General → About this build for API base.
2. Confirm Register #1 is not pointed at `127.0.0.1`.
3. Open the Backoffice / Server PC URL from Register #1.
4. Run Repair from Deployment Manager if station config is wrong.

PostgreSQL is down or unreachable

1. Open the Deployment Manager.
2. Check the **PostgreSQL Status** panel.
3. If service shows **stopped**, click **Start PG**.
4. If connection shows **Failed**, verify PostgreSQL is installed and the service is running.
5. If `psql.exe` is not found, install PostgreSQL or set `psqlPath` in the config.
6. Use **Restart PG** if the service is running but connections are failing.

Receipt will not print

1. Check Printers & Scanners.
2. For network mode, confirm Epson IP and port `9100`.
3. For installed-printer mode, confirm Windows test page works.
4. Restart the desktop app after printer/network changes.
5. For iPad, confirm the server can reach the network printer.

Counterpoint bridge fails

1. Confirm `R0S_BASE_URL=http://SERVER-IP:3000`.
2. If bridge shows `health 503`, configure the token in Riverside or run `Repair-RiversideCredentialsKey.cmd`.
3. If bridge shows `health 401`, run `Set-CounterpointBridgeToken.cmd` on the server PC and paste the exact token from the bridge `.env`.
4. Restart the bridge after changing `.env`.

Update incomplete

1. Finish updating the Backoffice / Server PC.
2. Finish updating Register #1 desktop app.
3. On iPad/PWA, Check app files and reload.
4. Do not continue production work until all surfaces show `0.70.2`.

13. Deployment Log

Item	Value
Deployment date	
Release version	0.70.2
Backoffice / Server PC name	
Backoffice / Server PC IP	
Server API host on server PC	http://127.0.0.1:3000
Register #1 PC name	
Register #1 API base	
iPad Register #2 URL	
Counterpoint PC name	
Counterpoint ROS_BASE_URL	
Epson receipt printer IP/name	
Zebra scanner model(s)	
Backup file name	
Backup timestamp	
Installer owner	
Store manager signoff	
Rollback authority	